

Use Case Descriptions

Web-Based Vehicle Auction Platform

1. Actors

Actor	Description
Guest User	Unregistered user browsing the platform
Registered User	User with an account but not yet verified
Verified Buyer	Admin-approved user allowed to bid
Admin	System administrator with full privileges

2. Use Case List

Use Case ID	Use Case Name	Primary Actor
UC-01	Register Account	Guest User
UC-02	Login	Registered User
UC-03	Browse Vehicle Listings	Guest User
UC-04	View Vehicle Details	Guest User
UC-05	Verify User Account	Admin
UC-06	Create Auction	Admin
UC-07	Place Bid	Verified Buyer
UC-08	View Bid History	Admin
UC-09	Select Winning Bidder	Admin

3. Detailed Use Case Descriptions

UC-01: Register Account

- **Primary Actor:** Guest User
- **Description:** Allows a new user to create an account on the system.
- **Preconditions:** User is not logged in.
- **Postconditions:** User account is created with “Unverified” status.

Main Flow:

1. Guest selects “Register”.
2. System displays registration form.
3. User enters required details.
4. System validates input.
5. System creates user account.

Alternate Flow:

- 4a. Invalid data entered → system displays error message.

UC-02: Login

- **Primary Actor:** Registered User
- **Description:** Authenticates a user and grants system access.
- **Preconditions:** User has a registered account.
- **Postconditions:** User session is created.

Main Flow:

1. User enters credentials.
2. System validates credentials.
3. System logs the user in.

Alternate Flow:

- 2a. Invalid credentials → login denied.

UC-03: Browse Vehicle Listings

- **Primary Actor:** Guest User
- **Description:** Allows users to browse available vehicle auctions.
- **Preconditions:** System is operational.
- **Postconditions:** Vehicle list displayed.

Main Flow:

1. User accesses vehicle listing page.
2. System displays available vehicles.
3. User applies search or filters.

UC-04: View Vehicle Details

- **Primary Actor:** Guest User
- **Description:** Displays detailed information of a selected vehicle.
- **Preconditions:** Vehicle exists.
- **Postconditions:** Vehicle details displayed.

Main Flow:

1. User selects a vehicle.
2. System displays specifications, images, and auction status.

UC-05: Verify User Account

- **Primary Actor:** Admin
- **Description:** Admin approves or rejects user verification requests.
- **Preconditions:** User account exists.
- **Postconditions:** User status updated.

Main Flow:

1. Admin logs in.
2. Admin views pending users.
3. Admin approves or rejects user.
4. System updates user status.

UC-06: Create Auction

- **Primary Actor:** Admin
- **Description:** Admin creates an auction for a vehicle.
- **Preconditions:** Vehicle listing exists.
- **Postconditions:** Auction scheduled.

Main Flow:

1. Admin selects a vehicle.
2. Admin sets start and end time.
3. System validates timing.
4. Auction is created.

UC-07: Place Bid

- **Primary Actor:** Verified Buyer
- **Description:** Allows verified buyers to place bids on active auctions.
- **Preconditions:** User is verified; auction is active.
- **Postconditions:** Bid recorded.

Main Flow:

1. Buyer selects an active auction.
2. System displays current highest bid.
3. Buyer enters bid amount.
4. System validates bid.
5. System records bid.

Alternate Flows:

- 4a. Bid below minimum → rejected.
- 4b. Auction closed → rejected.

UC-08: View Bid History

- **Primary Actor:** Admin
- **Description:** Displays bidding history for an auction.
- **Preconditions:** Auction exists.
- **Postconditions:** Bid history displayed.

Main Flow:

1. Admin selects an auction.
2. System displays all bids with timestamps.

UC-09: Select Winning Bidder

- **Primary Actor:** Admin
- **Description:** Admin selects the winning bidder after auction ends.
- **Preconditions:** Auction is closed.
- **Postconditions:** Winning bidder recorded.

Main Flow:

1. Admin reviews highest bid.
2. Admin confirms winning bidder.
3. System marks auction as completed.